

Tianyu Tu

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EDUCATION

Nanjing University, School of Astronomy and Space Science <i>PhD in Astronomy (Astrophysics) Supervisor: Prof. Yang Chen</i>	Nanjing, China Sep. 2022 – Jun. 2027 (expected)
Laboratoire d'Astrophysique de Bordeaux <i>Visiting PhD student co-Supervisor: Dr. Valentine Wakelam</i>	Pessac, France Dec. 2024 – Dec. 2025
Nanjing University, Kuang Yaming Honors School <i>B.S. in Astronomy</i>	Nanjing, China Sep. 2018 – Jun. 2022

PUBLICATIONS

- V. Wakelam*, **T.-Y. Tu**, *Testing various assumptions for radiolysis, non-diffusive chemistry, and chemical desorption in cold cores*. A&A (2026).
- **T.-Y. Tu***, W. Yang, S. Feng, V. Wakelam, Y. Chen*, P. Zhou, Q.-Q. Zhang, *An ACA map of a molecular cloud interacting with supernova remnant W28*. ApJ, 1006, 242 (2026).
- **T.-Y. Tu***, A. Boogert, Y. Chen*, W. He, *Dust Absorption toward Supernova Remnant W44*. AJ, 171, 323 (2026).
- **T.-Y. Tu**, V. Wakelam*, J.-C. Loison, M. Chabot, E. Dartois, Y. Chen*, *Simulation of proton radiolysis of H₂O and O₂ ices with the Nautilus code*. A&A, 708, A237 (2026).
- S.-D. Bo, Y. Huang, P. Zhou*, **T.-Y. Tu**, S. Safi-Harb*, Z.-Y. Zhang, Y. Chen, and H. Sano, *JCMT ¹²CO J = 3–2 Observations of Tycho's Supernova Remnant: Constraints on the Environmental Gas Properties*. ApJ, 997, 154 (2026).
- Q.-Q. Zhang, Y. Chen*, Q.-C. Liu, X. Zhou, **T.-Y. Tu**, *A New Investigation of the Molecular Environment of Supernova Remnant G40.5–0.5*. AJ, 170, 305 (2025).
- Y.-Z. Shen, Y. Chen*, X. Zhang*, **T.-Y. Tu**, W.-J. Zhong, Q.-Q. Zhang, Q.-C. Liu, *Expanding Molecular Shell and Possible γ -ray Source Associated with Supernova Remnant Kesteven 67*. MNRAS, 537, 464 (2025).
- **T.-Y. Tu**, V. Wakelam, Y. Chen*, P. Zhou, Q.-Q. Zhang, *Molecular chemistry induced by J-shock toward supernova remnant W51C*. A&A, 693, A222 (2025).
- **T.-Y. Tu**, Y. Chen*, Q.-C. Liu, *Mapping the dense molecular gas towards thirteen supernova remnants*. ApJ, 978, 83 (2025).
- **T.-Y. Tu**, P. Rayalacheruvu, L. Majumdar, Y. Chen*, P. Zhou, M. Santander-García, *A Yebes W band Line Survey towards an Unshocked Molecular Cloud of Supernova Remnant 3C391: Evidence of Cosmic-Ray-Induced Chemistry*. ApJ, 974, 262 (2024).
- **T.-y. Tu**, Y. Chen*, P. Zhou, S. Safi-Harb., Q.-C. Liu, *Shock and Cosmic-Ray Chemistry Associated with the Supernova Remnant W28*. ApJ, 966, 178 (2024).

CONFERENCE CONTRIBUTIONS

Frontiers and Technology of Radio Astronomy Delingha, China Talk, <i>Observation of the physical and chemical feedback of supernova remnants on molecular clouds</i>	Aug. 2026
The 46th COSPAR Scientific Assembly 2026 Florence, Italy Talk, <i>Simulation of proton radiolysis of H₂O and O₂ ices with the Nautilus code</i> Poster, <i>Molecular chemistry induced by a J-shock toward supernova remnant W51C</i> Poster, <i>An ACA map of a molecular cloud interacting with supernova remnant W28</i>	Aug. 2026
Laboratory atomic and molecular astrophysics and molecular astrochemistry Tianshui, China Talk, <i>Chemical feedback of supernova remnants: dust absorption toward SNR W44</i>	Jul. 2026
Astrochemistry: From Quantum to Cosmos Nijmegen, the Netherlands Talk, <i>Simulation of proton radiolysis of H₂O and O₂ ices with Nautilus</i>	Oct. 2025
Cosmic Rays 3: The salt of the star formation recipe Florence, Italy Talk, <i>Chemistry induced by cosmic rays towards supernova remnants W28 and 3C391</i>	Oct. 2024

Supernova Remnant III: An Odyssey in Space after Stellar Death Chania, Greece	Jun. 2024
Poster, <i>A Yebes W band line survey towards Supernova remnant 3C391: Evidence of CR-induced chemistry</i>	
Molecular Astrochemistry Workshop Urumchi, China	Aug. 2023
Talk, <i>A Yebes W band line survey towards Supernova remnant 3C391: Evidence of CR-induced chemistry</i>	
The 10th Physics and Chemistry in the Interstellar Medium Shijiazhuang, China	Jul. 2023
Talk, <i>Shock and Cosmic Ray Chemistry in Supernova Remnant W28</i>	
Supernova Remnant and Interstellar Medium 2023 Suzhou, China	Jul. 2023
Talk, <i>Shock and Cosmic Ray Chemistry in Supernova Remnant W28</i>	

APPROVED OBSERVING PROPOSALS

PI, ALMA, 2026.1.00381.S, 7m: 42.9 h, TP: 73.5 h	2026
<i>Mapping CH₃OH and H₂S in starless cores</i>	
PI, Gemini South/Flamingos2, GS-2026A-Q-334, 2.92 h	2025
<i>Investigating the origin of the H₂ emission toward supernova remnant RCW 86</i>	
PI, GBT, GBT26A-008, 39.25 h	2025
<i>NH₃ map of molecular clouds interacting with supernova remnants W28 and IC443</i>	
PI, IRAM 30m, 168-25, 19.0 h	2025
<i>Shock and cosmic-ray chemistry induced by supernova remnant W28</i>	
PI, IRAM 30m, 135-25, 34.0 h	2025
<i>Complex organic molecules in starless cores: a sample in the California molecular cloud</i>	
PI, ALMA, 2025.1.00086.S, 7m: 35.4 h, TP: 58.6 h	2025
<i>Is SO⁺ a tracer of dissociative shock in supernova remnant IC443?</i>	
PI, CTIO-4m/NEWFIRM, 2025B-571615, 1.5 nights	2025
<i>Probing the shock-cloud interaction toward superbubble 30 Doradus C</i>	
PI, Gemini North/GNIRS-IFU, GN-2025B-Q-143, 5.86 h	2025
<i>Probing the farthest supernova remnant interacting with molecular cloud with H₂</i>	
PI, P200/WIRC, CTAP2025-B0022, 1 night (through TAP)	2025
<i>Is the Tycho's supernova remnant interacting with ambient molecular clouds?</i>	
PI, IRAM 30m, 006-25, 28.9 h	2025
<i>Cosmic ray ionization towards supernova remnant HB9</i>	
PI, IRAM 30m, 016-25, 39.4 h	2025
<i>Origin of the narrow SiO emission in IRDC G33.69-0.01</i>	
PI, NOEMA, S25AN, 19.0 h	2025
<i>J- and C-shock chemistry towards clump C of supernova remnant IC443</i>	
PI, NOEMA, S25AS, 15.6 h	2025
<i>Probing the molecular cloud exposed to shock and cosmic rays from supernova remnant 3C391</i>	
PI, Yebes, 25A004, 27.2 h	2025
<i>A Q-band line survey towards supernova remnant W28</i>	
PI, ATCA (H75 configuration), C3530 (re-submission), 5.5 h	2024
<i>NH₃ observation towards three ATLASGAL clumps associated with supernova remnant W28</i>	
PI, IRTF, 2025A003, 28.2 h	2024
<i>Cosmic-ray processing of interstellar ice towards supernova remnant W51C</i>	
PI, IRTF, 2025A007, 14 h	2024
<i>Probing the cosmic-ray ionization towards the Cygnus Cocoon</i>	
PI, CTIO-4m/NEWFIRM, 2025A-734415, 2 nights	2024
<i>Probing the interaction between supernova remnants and molecular clouds towards TeV gamma-ray sources</i>	
PI, CTIO-4m/NEWFIRM, 2025A-127790, 0.5 nights	2024
<i>Probing the shock-cloud interaction towards supernova remnant RX J1713.7-3946</i>	
PI, CFHT/WIRCAM, 25AS01, 4.54 h (through TAP)	2024
<i>Probing the interaction between supernova remnant and molecular cloud towards Galactic PeVatron: G106.3+2.7</i>	

PI, ALMA, 2024.1.00194.S, 7m: 11.6 h, TP: 23 h <i>Physics and chemistry of dense molecular gas interacting with SNR W28</i>	2024
PI, IRTF, 2024B008, 11 h <i>Cosmic-ray processing of interstellar ice towards supernova remnant W44</i>	2024
PI, JCMT, M24BP015, 23.7 h, filler <i>Estimation of cosmic-ray ionization rate towards supernova remnant Kes79</i>	2024
PI, Yebes, 24B010, 36.5 h <i>Chemistry induced by cosmic rays towards supernova remnant Kes79</i>	2024
PI, Yebes, 24A003, 30.5 h <i>Molecular Chemistry Induced by J-shock in Supernova Remnant W51C</i>	2024
PI, ATCA (H168+H75 configurations), C3530 (re-submission), 10.6 h <i>NH₃ observation towards three ATLASGAL clumps associated with supernova remnant W28</i>	2023
PI, GBT, GBT23B-090, 12 h <i>Probing the cosmic-ray-ionized dense gas associated with supernova remnant HB9</i>	2023
PI, Yebes, 23A021, 22 h <i>A 3mm line survey towards supernova remnant 3C391</i>	2023
PI, ATCA (H168 configuration), C3530, 24 h <i>NH₃ observation towards three ATLASGAL clumps associated with supernova remnant W28</i>	2022
PI, SMA (compact configuration), 2022B-S009, 16 h <i>Shock and cosmic ray chemistry in molecular clouds interacting with supernova remnant W28</i>	2022
PI, NOEMA (D configuration), W22AS, 11 h <i>Shock and cosmic ray chemistry in molecular clouds interacting with supernova remnant W28</i>	2022
co-I, PMOD, 39 h PI: Yang Chen	2026
co-I, PMOD, 65 h PI: Yang Chen	2024
co-I, IRAM 30m, 17.6 h PI: Qianqian Zhang	2024
co-I, PMOD, 60 h PI: Yang Chen	2024
co-I, PMOD, 79 h PI: Yang Chen	2022
co-I, PMOD, 58 h PI: Yang Chen	2022
ADDITIONAL EXPERIENCE	
Teaching Assistant	
Nanjing University, School of Astronomy and Space Science	Sep. 2023 – Jan. 2024
Course for the graduate: <i>Radiation Mechanisms in Astrophysics</i>	
Leadership/Teamwork Experience	
Great Astronomy Fans Association of Nanjing University <i>President</i>	Jun. 2019 – Jun. 2020
Astronomy Union of Nanjing Universities <i>Minister</i>	Mar. 2019 – Mar. 2020
HONORS & AWARDS	
Daixia Scholarship (top ~ 10%) <i>Nanjing University</i>	2025
Outstanding graduate student <i>Nanjing University</i>	2024, 2025
Academic Scholarship (Second Class, top ~ 40%) <i>Nanjing University</i>	2021
Wanglaoji Scholarship (top ~ 10%) <i>Nanjing University</i>	2020
Academic Scholarship (First Class, top ~ 20%) <i>Nanjing University</i>	2019, 2020
SKILLS & INTERESTS	
Languages: Mandarin Chinese (native), English (fluent)	
Programming: Python, Unix Shell (basic), Fortran (basic)	
Observing experience: ATCA (remote), IRTF/SpeX & iSHELL (remote), Blanco-4m/NEWFIRM (remote), IRAM 30m (on-site & remote), P200/WIRC (remote), GBT (remote)	
Data reduction softwares Gildas, Miriad, CASA, Spextool, IRAF	
Codes: Nautilus, Paris-Duram shock code, UCLCHEM, Meudon PDR code, PLUTO, RADEX, etc.	
Academic Interests: Molecular chemistry, Supernova remnants and their feedback	
Invited referee for ApJ	